

MIS-04003 ABBREVIATIONS AND SYMBOLS FOR PFD AND P & ID

1 General

1.1 Scope

This specification gives some guidance to understand process flow diagram (PFD) and piping and instrument diagram (P & ID).

1.2 Numbering System

Followings are described in "BASIC ENGINEERING DATA".

- (a) Equipment numbering system
- (b) Instrument numbering system

2 Equipment

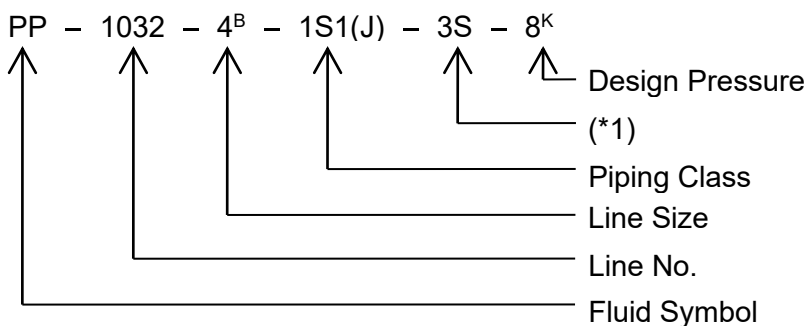
Graphic symbols for equipment are shown in TABLE-1 "GRAPHIC SYMBOLS FOR EQUIPMENT".

Outline sketches of equipment may be used instead of graphic symbols in TABLE-1 if required.

3 Pipe Line

3.1 Line Number

Line numbers are shown as follows.



(*1)

H	:	Hot Insulation	C	:	Cold Insulation
P	:	Personnel Protection	D	:	Anti-sweating
25S	:	25 ^K Steam Tracing + Insulation			
10S	:	10 ^K Steam Tracing + Insulation			
3S	:	3 ^K Steam Tracing + Insulation			
S	:	1 ^K Steam Tracing + Insulation			
HO	:	Hot Oil Tracing + Insulation			
PN	:	Noise Protection			

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3.2 Line Symbols

Line symbols for PFD and P & ID are shown in TABLE-2 “LINE SYMBOLS”.

3.3 Fluid Symbols

Fluid Symbols for PFD are shown in TABLE-3 “FLUID SYMBOLS FOR PFD”.

3.4 Fluid Service Symbols

Fluid symbols for P & ID are shown in TABLE-4 “FLUID SYMBOLS FOR P & ID”.

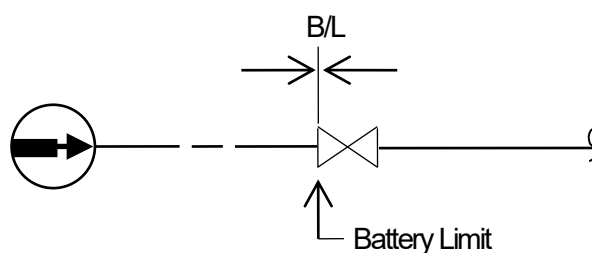
4 Fitting and Piping Accessories

Letters and graphic symbols for fittings and piping accessories are shown in TABLE-5 “LETTERS AND GRAPHIC SYMBOLS FOR FITTINGS AND PIPING ACCESSORIES”.

5 Miscellaneous Symbols

5.1 Battery Limit Mark and Connection Style

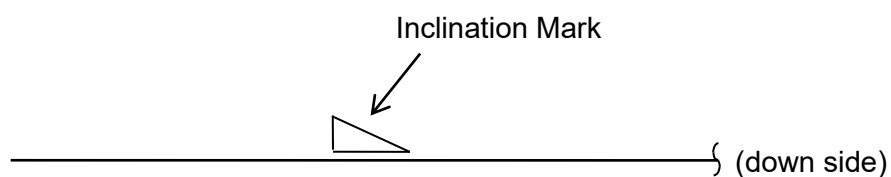
The battery limit mark shall be shown on the point where materials or products are receiving or delivering. In case that the pipe lines are connected with lines outside the battery limit, connection styles shall be shown as follows.



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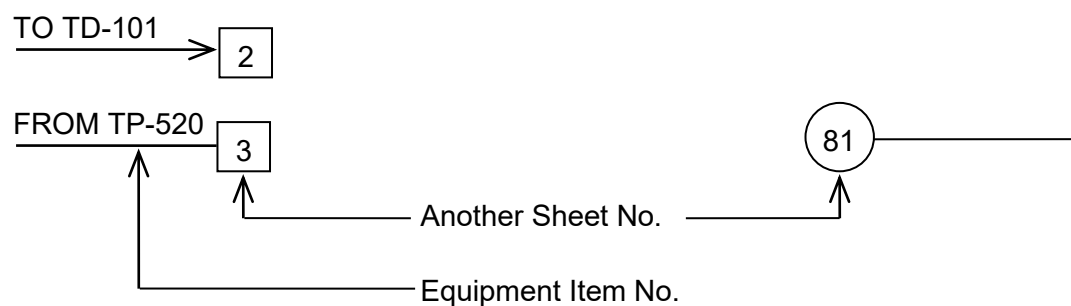
5.2 Inclination Mark

Inclination shall be shown upside of the line to be inclined in the following manner.



5.3 Connection Mark to Another Sheet

Connection to another sheet shall be shown as follows.

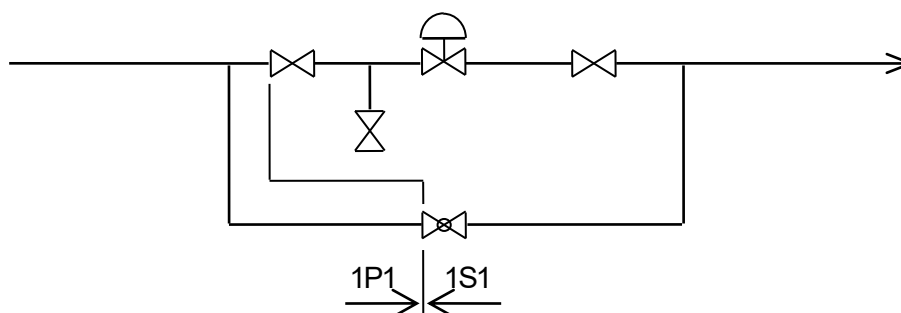


□ indicates connection to or from process P & ID

○ indicates connection to or from utility P & ID

5.4 Split Mark of Piping Material Classification

Piping class split shall be shown as follows.



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5.5 Hot, Cold, and Personnel Protection Insulation

Hot, cold, and Personnel Protection Insulation are shown in TABLE-17.

5.6 Others

Following abbreviations shall be used on the P & ID.

<u>Symbols</u>	<u>Descriptions</u>
IH	Inspection Hole
HH	Hand Hole
MH	Manhole
MAX	Maximum
MIN	Minimum
W/E	Supplied with Equipment
W/I	Supplied with Instrument
C.S.O	Car Seal Open
C.S.C	Car Seal Close
SP	Set Pressure
NPT	Pipe Thread

6 Instrument

6.1 Instrument Symbols

6.1.1 TABLE-6 covers the permissible combinations of letters of identification and shows the significance of such identification.

6.1.2 Basic graphic symbols for instruments and instrumentation are shown in TABLE-7 and its application examples are shown in TABLE-8 through 12.

6.1.3 TABLE-8 through 12 show how to designate instruments and their elements on the P & ID.

6.1.4 A brief explanatory notation may be added alongside the symbol, when necessary to clarify the function or purpose of the instrument.

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6.1.5 Purge system for FT (Flow transmitter), LT (Level transmitter) and PT (Pressure transmitter) are shown in TABLE-13, 14 and 15.

6.2 Additional Notes

6.2.1 Failure Position

The Power failure position of all control valves including emergency shut-off valves and sequence control valves shall be indicated on the P & ID using the symbols of FC, FO, FL and FH. The meanings of these symbols are as follows.

FC : Closed when power failure
FO : Opened when power failure
FL : Locked when power failure
FH : Held when power failure

6.2.2 Valve Size and Type

Valve size and type using lead piping are shown on TABLE-16.

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TABLE-1 GRAPHIC SYMBOLS FOR EQUIPMENT

<u>Descriptions</u>	<u>Graphic Symbols</u>	
<u>Pumps</u>		
Centrifugal Pump or Canned Pump		
Displacement Pump (Rotary Pump, Reciprocating Pump, Gear Pump)		
<u>Compressors</u>		
Centrifugal Blower or Fan		
Rotary Blower or Compressor		
Reciprocating Compressor		
Centrifugal or Axial Compressor		
<u>Drivers</u>		
Electric Motor		
Turbine		

TABLE-1 GRAPHIC SYMBOLS FOR EQUIPMENT (Cont'd)

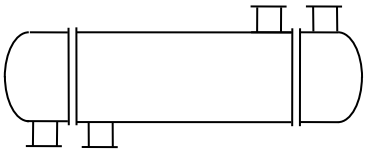
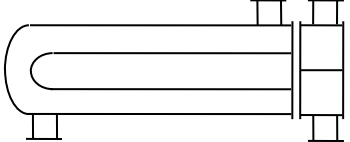
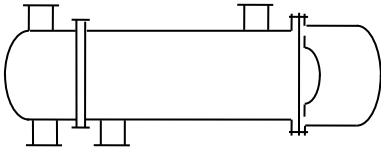
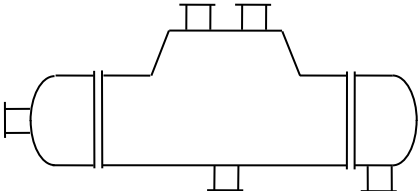
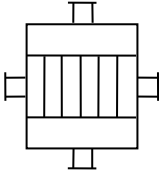
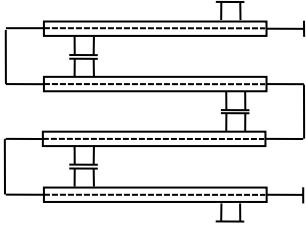
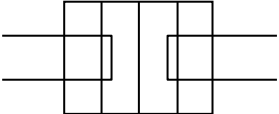
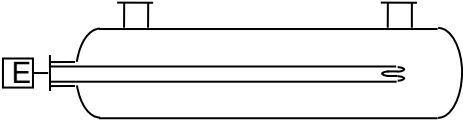
<u>Descriptions</u>	<u>Graphic Symbols</u>
<u>Heat Exchangers</u>	
Fixed Tube Type Exchanger	
U Tube Type Exchanger	
Floating Head Type Exchanger	
Kettle Type Exchanger	
Spiral Type Exchanger	
Double Pipe Exchanger	
Plate Type Exchanger	
Electric Heater	

TABLE-1 GRAPHIC SYMBOLS FOR EQUIPMENT (Cont'd)

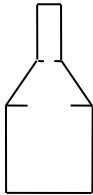
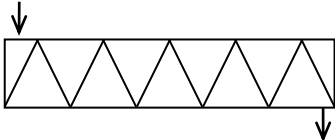
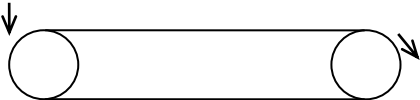
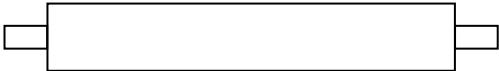
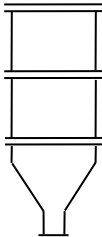
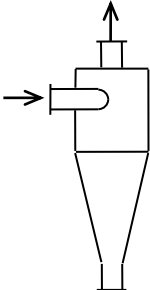
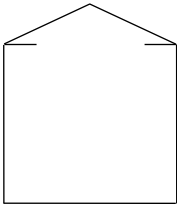
<u>Descriptions</u>	<u>Graphic Symbols</u>
<u>Furnace</u>	
Heater	
<u>Conveyors</u>	
Screw Conveyor	
Band, Belt, Shaker Conveyor	
<u>Others</u>	
Dryer	
Centrifuge	
Cyclone	
Drum	 Vertical  Horizontal
Cone Roof Tank	

TABLE-1 GRAPHIC SYMBOLS FOR EQUIPMENT (Cont'd)

Descriptions

Graphic Symbols

Silo

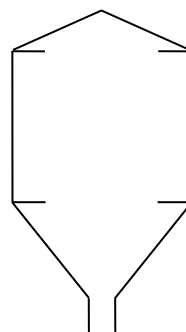
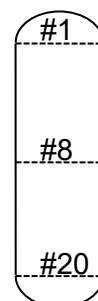
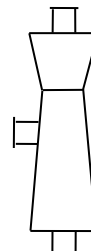


Plate Tower



Desuperheater



Packed Tower

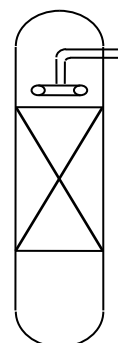
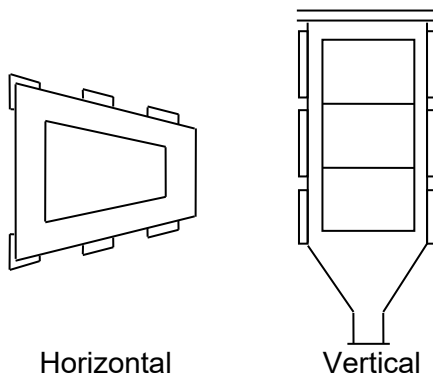


TABLE-1 GRAPHIC SYMBOLS FOR EQUIPMENT (Cont'd)

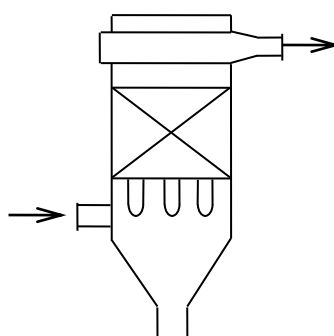
Descriptions

Graphic Symbols

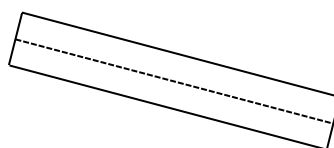
Film Evaporator



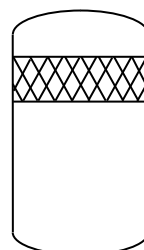
Bag Filter



Vibrating Screen



Demister



Spray Nozzle

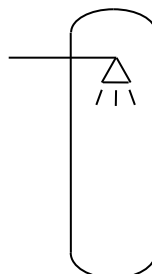
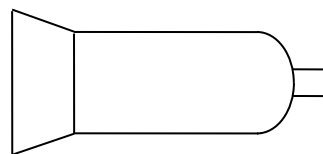


TABLE-1 GRAPHIC SYMBOLS FOR EQUIPMENT (Cont'd)

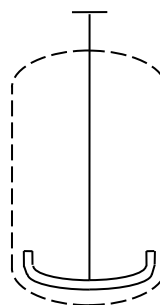
Descriptions

Graphic Symbols

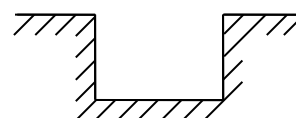
Silencer



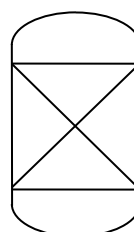
Agitator



Pit



Packed Bed Drum



Heating Coil

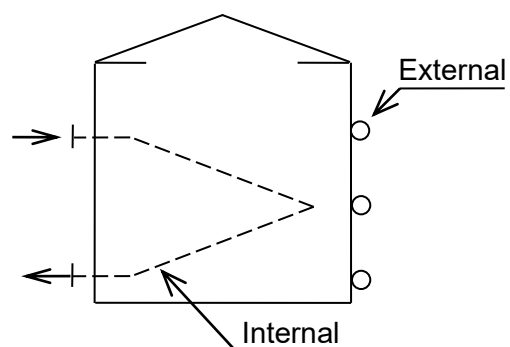


TABLE-2 LINE SYMBOLS

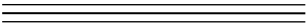
<u>Descriptions</u>	<u>Graphic Symbols</u>
Main Process Line	 Gross Solid
Other Process Line & Utility Line	 Medium Solid
Line to be shown on the Other Sheets & Under Ground Line	 Medium Broken
Traced Line	
Jacketed Line	
Line outside the Battery Limit	

TABLE-3 FLUID SYMBOLS FOR PFD

<u>Symbols</u>	<u>Services</u>
TA	Terephthalic Acid
PTA	Purified Terephthalic Acid
PX	Para-Xylene
Q	Acetic Acid
FDQ	Frash Distillated Acetic Acid
RQ	Recycle Acetic Acid
DQ	Dilute Acetic Acid
FQ	Fresh Acetic Acid
ML	Mother Liquor
WL	Wash Liquor
WG	Waste Gas
DIW	Deionized Water
LPW	Low Pressure Seal & Flushing Water
MPW	Medium Pressure Seal & Flushing Water
HPW	High Pressure Seal & Flushing Water
CCW	Circulating Cooling Water Supply
RCCW	Circulating Cooling Water Return
FLW	Filtrated Water
25S	25 kg/cm ² G Steam
15S	15 kg/cm ² G Steam
10S	10 kg/cm ² G Steam
5S	5 kg/cm ² G Steam
3S	3 kg/cm ² G Steam
1.7S	1.7 kg/cm ² G Steam
1S	1 kg/cm ² G Steam
-0.3S	-0.3 kg/cm ² G Steam
25SC	25 kg/cm ² G Steam Condensate
15SC	15 kg/cm ² G Steam Condensate
10SC	10 kg/cm ² G Steam Condensate
5SC	5 kg/cm ² G Steam Condensate
3SC	3 kg/cm ² G Steam Condensate
1SC	1 kg/cm ² G Steam Condensate
10BW	Pressurized Boiler Water (10kg/cm ² G)
3BW	Pressurized Boiler Water (3kg/cm ² G)
1BW	Pressurized Boiler Water (1kg/cm ² G)
FO	Fuel Oil
HO	Hot Oil
7N	7 ^K Nitrogen
9N	9 ^K Nitrogen
15N	15 ^K Nitrogen
WW(C)	Waste Water (Clean)
WW(R)	Waste Water (Rain)
WW(B)	Waste Water (Batch-wise)
WW(P)	Waste Water (Process)

TABLE-4 FLUID SYMBOLS FOR P&ID

<u>Symbols</u>	<u>Services</u>
TP	TA UNIT Process Fluid
PA	Process Air
HFQ	High Pressure Seal & Flushing Acetic
Acid	
FDQ	Flash Distillated Acetic Acid
FQ	Fresh Acetic Acid
PP	PTA UNIT Process Fluid
H	Hydrogen
LPW	Low Pressure Seal & Flushing Water
MPW	Medium Pressure Seal & Flushing Water
HPW	High Pressure Seal & Flushing Water
10BW	Pressurized Boiler Water (10kg/cm ² G)
3BW	Pressurized Boiler Water (3kg/cm ² G)
1BW	Pressurized Boiler Water (1kg/cm ² G)
FLW	Filtrated Water
CCW	Circulating Cooling Water Supply
RCCW	Circulating Cooling Water Return
PW	Potable Water
DIW	Deionized Water
IA	Instrument Air
SA	Service Air
25S	25 kg/cm ² G Steam
15S	15 kg/cm ² G Steam
10S	10 kg/cm ² G Steam
5S	5 kg/cm ² G Steam
3S	3 kg/cm ² G Steam
1.7S	1.7 kg/cm ² G Steam
1S	1 kg/cm ² G Steam
-0.3S	-0.3 kg/cm ² G Steam
25SC	25 kg/cm ² G Steam Condensate
15SC	15 kg/cm ² G Steam Condensate
10SC	10 kg/cm ² G Steam Condensate
5SC	5 kg/cm ² G Steam Condensate
3SC	3 kg/cm ² G Steam Condensate
1SC	1 kg/cm ² G Steam Condensate
FG	Fuel Gas
FO	Fuel Oil
7N	7 ^K Nitrogen
9N	9 ^K Nitrogen
15N	15 ^K Nitrogen
HN	High Pressure Nitrogen
WG	Waste Gas
FLG	Flare Gas
LO	Lube Oil
50NA	50% Caustic Soda
NA	3% Caustic Soda
SF	Sodium Formate

TABLE-4 FLUID SYMBOLS FOR P&ID (Cont'd)

<u>Symbols</u>	<u>Services</u>
HO	Hot Oil
WW(C)	Waste Water (Clean)
WW(R)	Waste Water (Rain)
WW(B)	Waste Water (Batch-wise)
WW(P)	Waste Water (Process)

TABLE-5 LETTERS AND GRAPHIC SYMBOLS FOR FITTING
AND PIPING ACCESSORIES

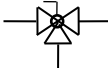
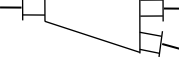
<u>Descriptions</u>	<u>Letter Symbols</u>	<u>Graphic Symbols</u>
Gate Valve	VG	
Globe Valve	VO	
Check Valve	VC	
Angle Valve	VA	
Butterfly Valve	VF	
L-Type Ball 3-way Valve	VBT	
Y-Type Ball 3-way Valve	VBY	
Slurry Drain Valve	VBE	
Ball Valve	VB	
Slurry Drain Bottom Flash Valve	VBFE	
Bottom Flash Valve	VBF	
Plug Valve (Cock)	VP	
Needle Valve	VN	
Diaphragm Valve	VD	
Jacket Type Gate Valve	JVG	
Safety Valve or Relief Valve	SV	
Breather Valve	BV	
Foot Valve	VFT	
Rotary Valve		
Two-way Valve		
Slide Gate Valve		
Jacket Type Ball Valve	JVB	
Jacket Y-Type Ball 3-way Valve	JVBY	
Pressure Relief Valve	PRV	

TABLE-5 LETTERS AND GRAPHIC SYMBOLS FOR FITTING
AND PIPING ACCESSORIES (Cont'd)

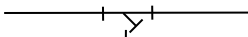
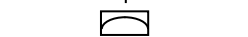
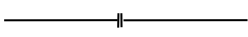
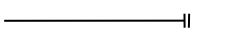
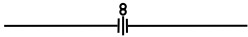
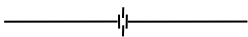
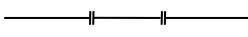
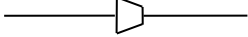
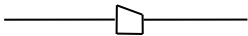
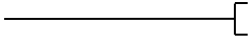
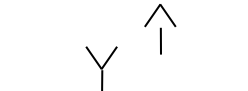
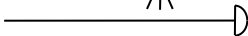
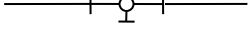
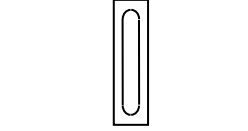
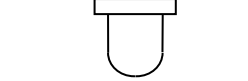
<u>Descriptions</u>	<u>Letter Symbols</u>	<u>Graphic Symbols</u>
Steam or Air Trap	ST/AT	
Basket Type Strainer	SF	
Y-type Strainer	SF	
Flame Arrester	FA	
Rupture Disc	RD	
Sight Glass	SG	
Flange		
Blind Flange		
Spade Flange	SP	
Spectacle Flange	SP	
Restriction Orifice	RO	
Flexible Hose	FX	
Flexible Tube		
Expansion Joint	EX	
Spool Piece		
Reducer (Concentric)	RC	
Reducer (Eccentric)	RE	
Hose Connection	HC	
Rain Hood		
Funnel/ Sewer		
Drainage to Pit		
Cap		
Sampling		
T-type Strainer	SF	
Measuring Gage	MG	
Cartridge Filter		

TABLE-5 LETTERS AND GRAPHIC SYMBOLS FOR FITTING
AND PIPING ACCESSORIES (Cont'd)

<u>Descriptions</u>	<u>Letter Symbols</u>	<u>Graphic Symbols</u>
Diaphragm Air Actuated Valve		
Electric Motor Operated Valve		
Solenoid Valve		
Hydraulically Operated Valve		
Pneumatically Operated Valve		
Gate Valve with Bypass Valve	VG	
Standard Elbow		
Bender Elbow		
Mixer	PM	
Line Mixer		
Vortex Breaker		
Temporary Strainer	TS	
Melting Plug		
Flexible Ball Joint	BJ	
Union or Coupling		
Plug		
Speed Controller for LO Unit		
Pressure Relief Valve for Pressure Unit		
Four (4) Way Solenoid Valve for Pressure Unit		

TABLE-6 INSTRUMENT SYMBOL

(a)DCS LOOP

x x x x x x

PROCESS VARIABLE

A: Analyzer
 F: Flow
 H: Hand
 I : Current
 xJ: Direct Input to DCS
 (x: Process Variables from A to Z)
 L: Level
 M: Motor
 P: Pressure (PD: Differential Pressure)
 T: Temperature (TJ: Direct Input to DCS)
 V: Vibration
 W: Weight
 X: Unclassified
 Z: Torque Limitter

LOOP FUNCTION

(Combination of Symbols for Multi-function)

A: Alarm
 C: Control
 G: Gauge
 I : Indication
 K: Sequence
 L: Light
 Q: Totalize
 S: Interlock
 Z: Switch

FINAL CONTROL ELEMENT

V: Valve

(b)REMOTE OPERATED VALVE BY DCS

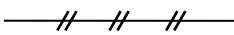
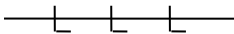
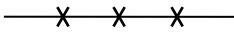
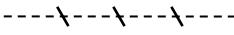
KCV: Sequence Control Valve
 SDV: Interlock Valve
 SKV: Sequence & Interlock Valve
 SOV: Solenoid Valve as KCV or SDV

(c)LOCAL INSTRUMENT

FG: Area Flow Meter
 LG: Level Gauge
 PG: Pressure Gauge
 PDG: Differential Pressure Gauge
 DG: Draft Gauge
 TG: Temperature Gauge
 HZ: Hand Operate
 PCV: Pressure Regulator

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TABLE-7 INSTRUMENT LINE SYMBOLS

Connection to Process or Mechanical Link, or Instrument Supply	
Pneumatic Signal	
Electric Signal	
Hydraulic Signal	
Capillary Tubing (Filled System)	
Electric Binary Signal	

Note: All lines to be finer than process lines.

General Instrument Symbols

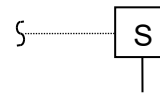
Locally Mounted	
Mounted on Console or Board	
Mounted on Local Board	
Distributed Control System with CRT (DCS)	

Actuator Symbols

Air Actuated Type for Control Valve	
Piston-Cylinder Type (Assembled in Process Equipments)	

TABLE-7 INSTRUMENT LINE SYMBOLS (Cont'd)

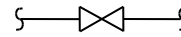
Solenoid Type



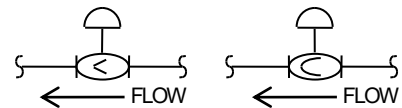
Note: Positioner and other accessories are not identified.

Control Valve Body Symbols

Globe, Gate, or Other in Line Type
not otherwise identified



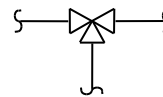
V-Ball, Eccentric Rotary



Angle Type



Three-way Type



Butterfly, Damper or Louver Type

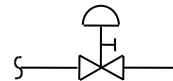


Symbols for Control Valve (Special Use)

Hand Control Valve



Control Valve with Hand Actuator



Pressure Reducing Regulator

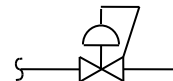
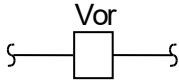
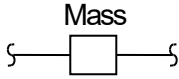
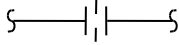
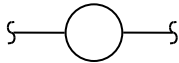
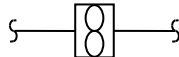
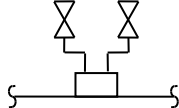


TABLE-7 INSTRUMENT LINE SYMBOLS (Cont'd)

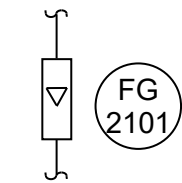
Symbols for Primary Elements

Vortex Flow	
Mass Flow	
Integral orifice	
Orifice Plate with Flange or Corner Taps	
Venturi Tube or Flow Nozzle	
Area Flow Meter	
Positive-Displacement Type	
Magnetic Flow Meter	
Turbine Type	
Positive Displacement Flow Meter	
Chemical Seal	
Annubar Type	

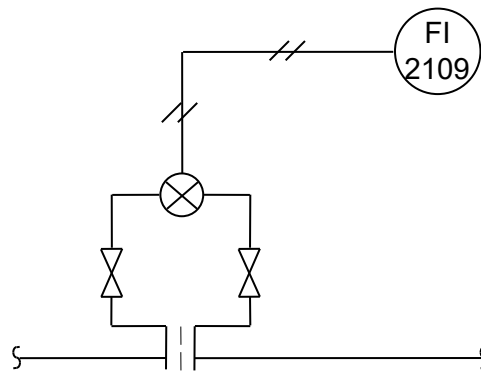
Symbols for Shutdown (Interlock)

SD- x x x x , Number Connected to shutdown Systems	SD- x x x x
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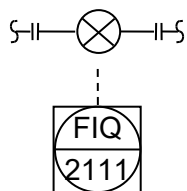
TABLE-8 TYPICAL INSTRUMENTATION SYMBOLS FOR FLOW



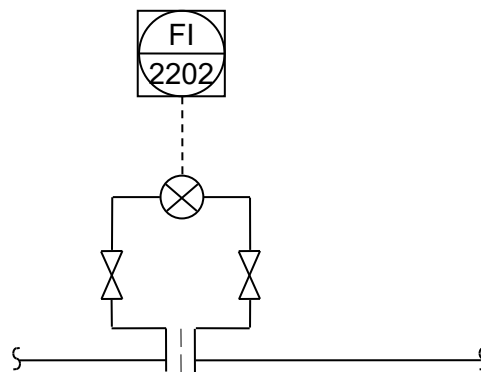
Rotameter



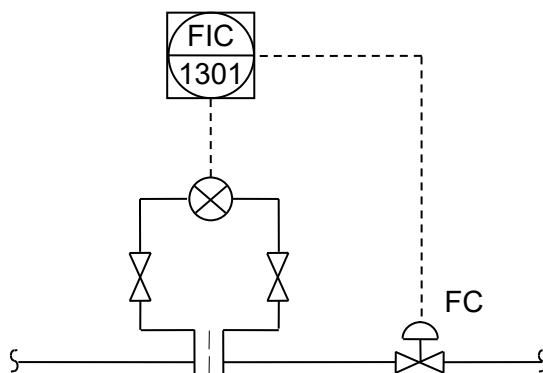
Orifice Plate with Flange Taps(Local)



Magnetic Flow Meter
Totalize by DCS

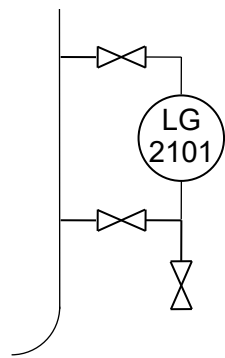


Differential Pressure Type
Flow Indicator by DCS

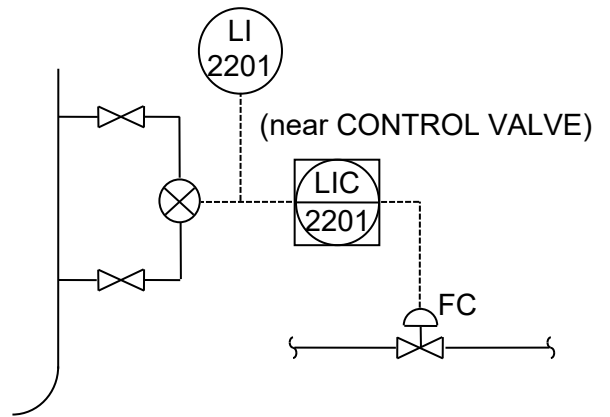


Flow Control by DCS

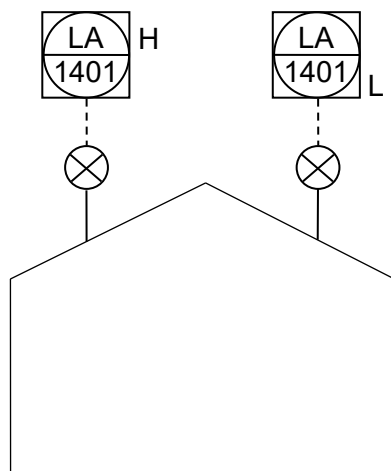
TABLE-9 TYPICAL INSTRUMENTATION SYMBOLS FOR LEVEL



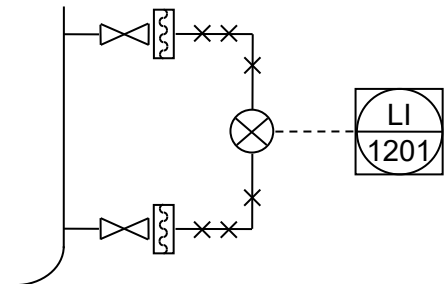
Level Gauge



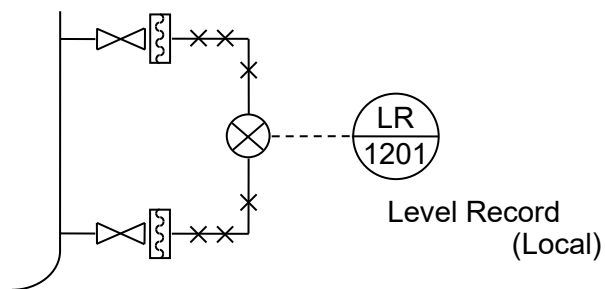
Level Control by DCS



Level Alarm by DCS

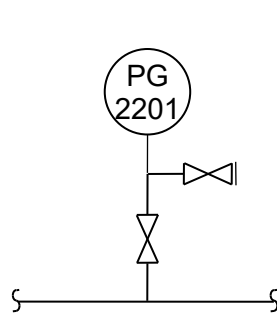


Differential Pressure Type
Level Transmitter Connected to
Diaphragm Seal
Level Indicator by DCS

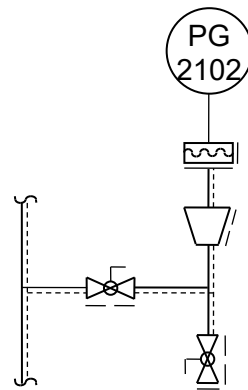


Level Record
(Local)

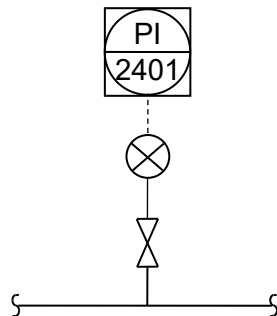
TABLE-10 TYPICAL INSTRUMENTATION SYMBOLS FOR PRESSURE



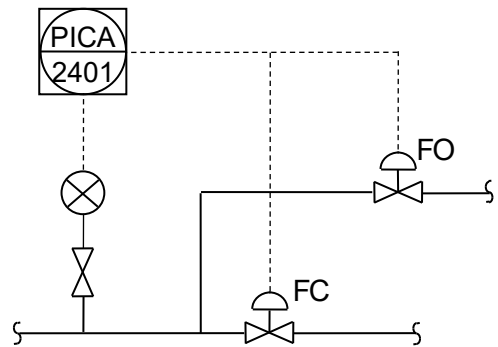
Direct Indicating
Pressure Gauge



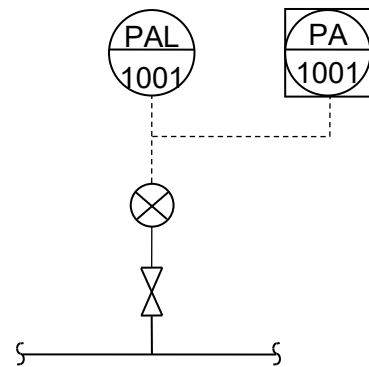
Direct Indicating
Pressure Gauge
with Chemical Seal



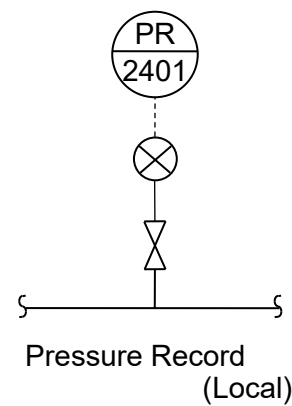
Pressure Indication by DCS



Pressure Control by DCS
Split Range I/P
Positioner is used



Pressure Switch
Pressure Alarm by DCS
and Hard wired Annunciator



Pressure Record
(Local)

TABLE-11 TYPICAL INSTRUMENTATION SYMBOLS FOR TEMPERATURE

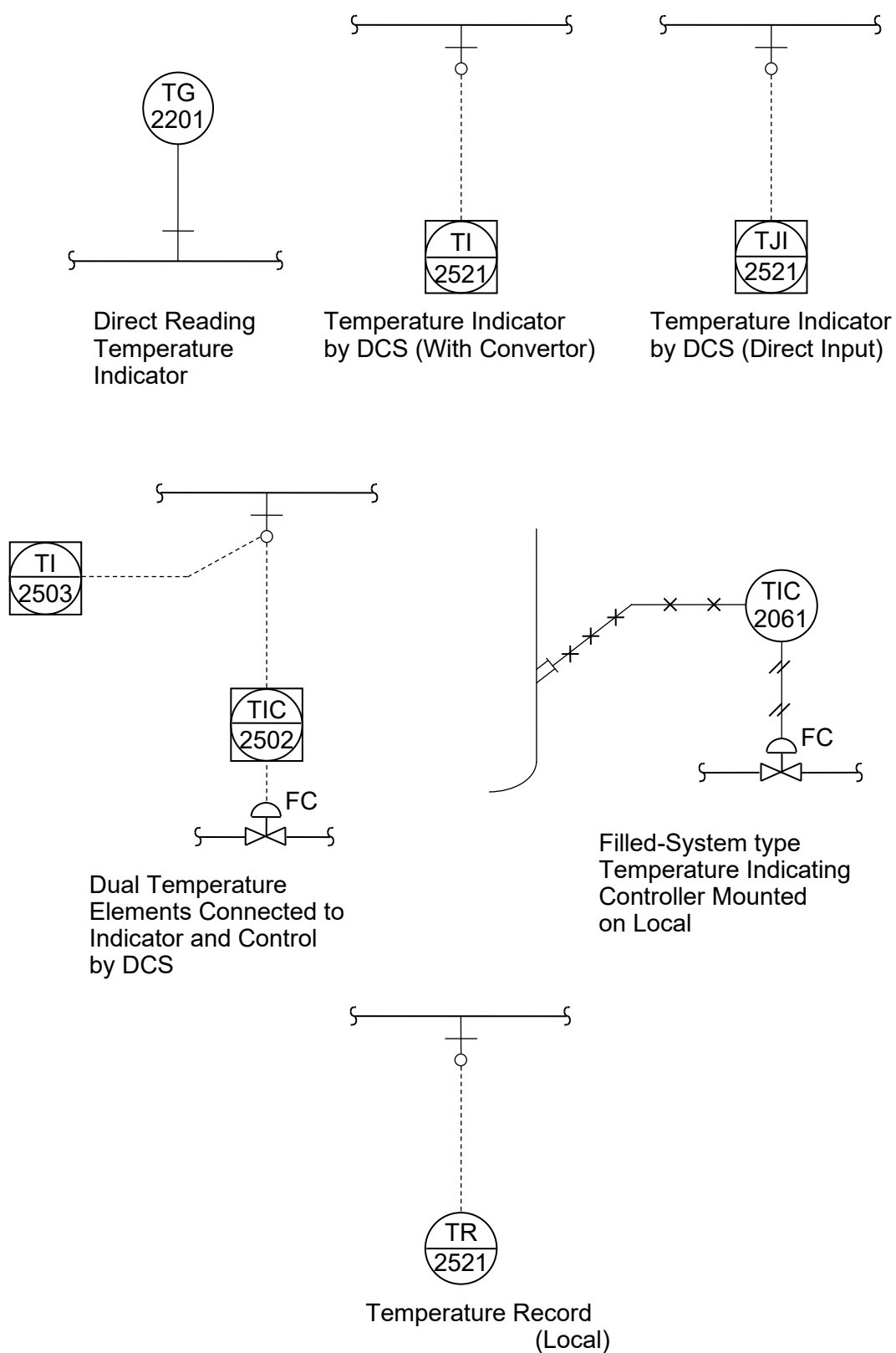


TABLE-12 TYPICAL INSTRUMENTATION SYMBOLS MISCELLANEOUS

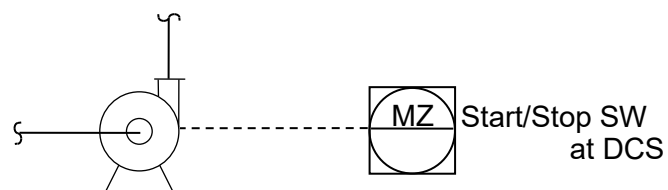
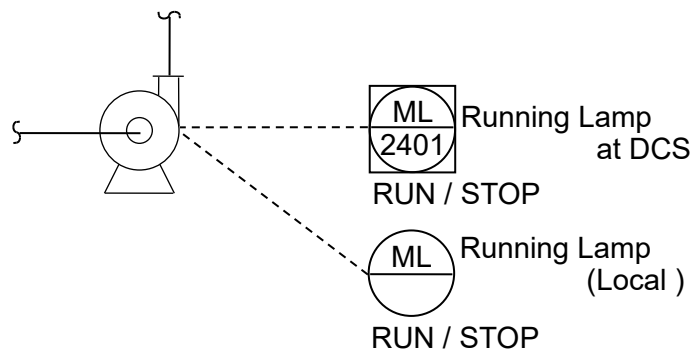
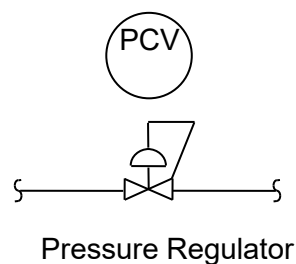
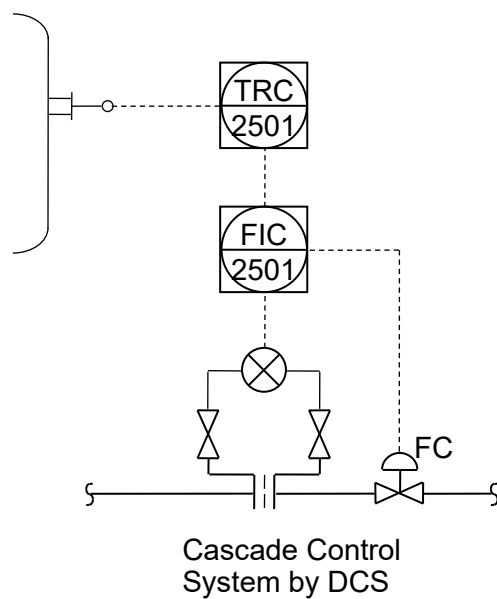
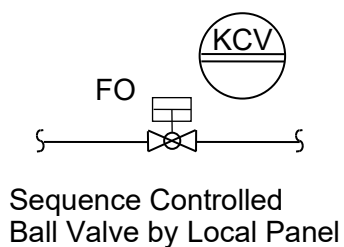
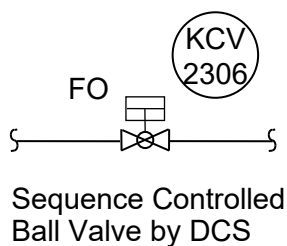
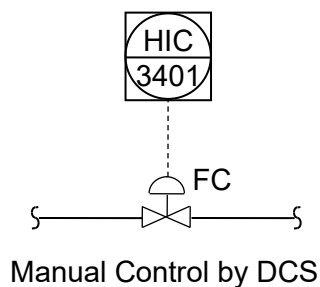
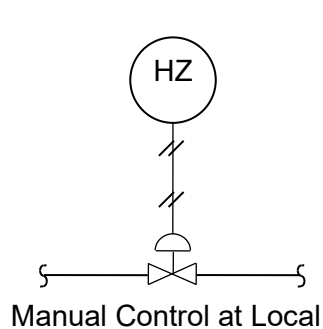
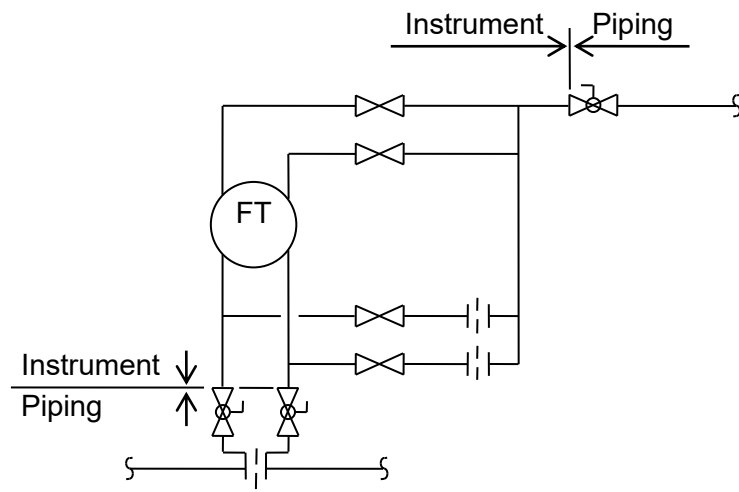


TABLE-13 PURGE SYSTEM FOR FT

Type "A"



Type "D"

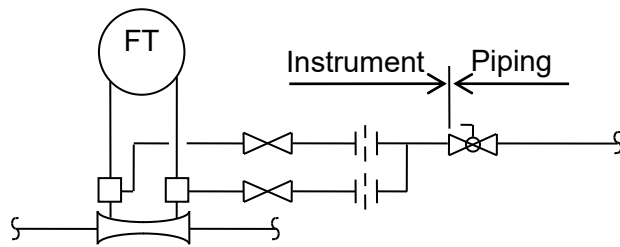
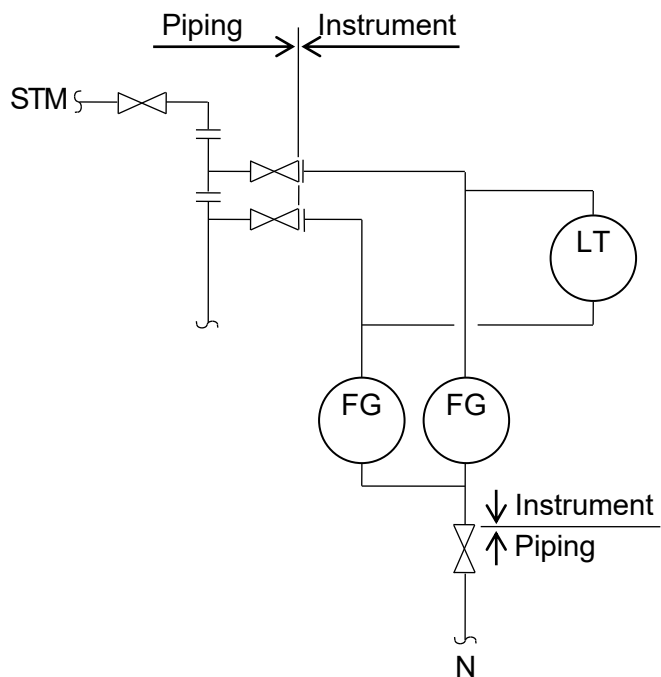
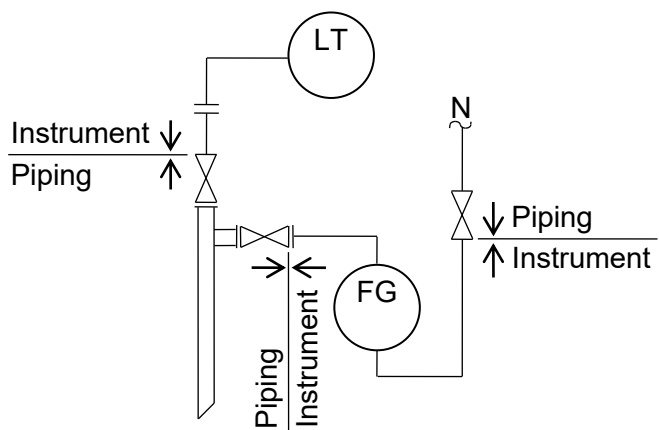


TABLE-14 PURGE SYSTEM FOR LT

Type "A"



Type "B"



Type "C"

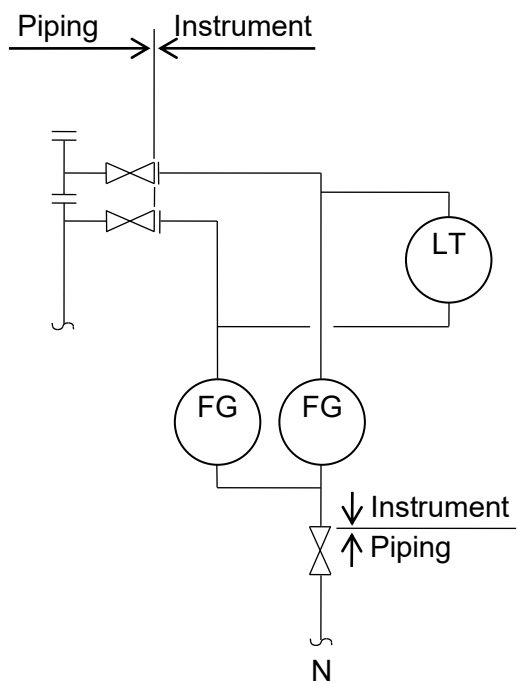


TABLE-15 PURGE SYSTEM FOR PT

Type "A"

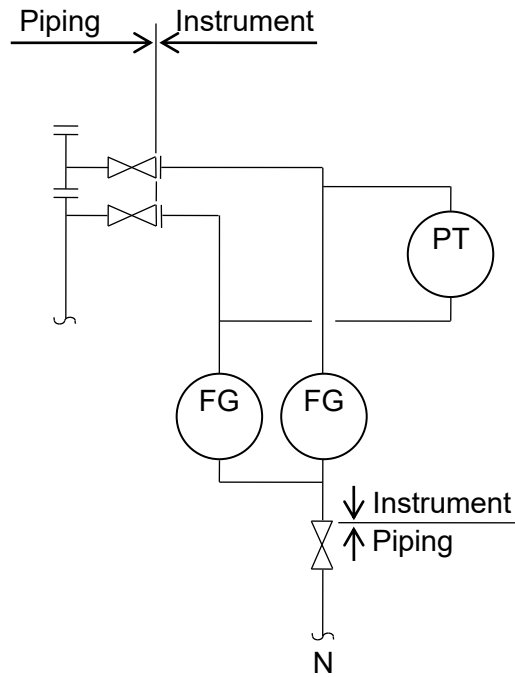


TABLE-16 VALVE SIZE AND TYPE

Valve size and type shall be as follows.

DESCRIPTIONS	STAINLESS STEEL	CARBON STEEL
Block valve connected to PG	3/4" Glove	3/4" Glove
Air release valve connected to PG	1/2" Ball	1/2" Gate
Valve connected to Orifice	1/2" Ball	1/2" Gate
Valve connected to PI	1/2" Ball	1/2" Gate
Valve connected to LIC	Specified on P&ID	
Drain valve connected to LI	1/2" Ball	1/2" Gate
Drain valve connected to LG	1/2" Ball	1/2" Gate
Drain valve connected to Strainer	3/4" Ball	3/4" Gate
Valve connected to Sampling	1/2" Ball	1/2" Gate
Drain valve connected to Piping or Equipment	3/4" Ball	3/4" Gate
Vent valve connected to Piping	1/2" Ball	1/2" Glove
Vent valve connected to Equipment or for Equipment	3/4" Ball	3/4" Glove
Vent valve connected to Jacket	3/4" Ball	3/4" Glove

TABLE-17 HOT, COLD, AND PERSONNEL PROTECTION INSULATION

Hot, cold, and personnel protection insulation dymbols shall be as follows.

Fluid Temperature		Hot Insulation Symbol
Below	124 °C	H
125 °C	~ 174 °C	H2
175 °C	~ 224 °C	H3
225 °C	~ 274 °C	H4
275 °C	~ 324 °C	H5
325 °C	~ 374 °C	H6
375 °C	~ 424 °C	H7
425 °C	~ 474 °C	H8
475 °C	~ 524 °C	H9
525 °C	~ 574 °C	H10

Fluid Temperature		Cold Insulation Symbol
Above	10 °C	C
9 °C	~ 0 °C	C2
-1 °C	~ -10 °C	C3
-11°C	~ -20 °C	C4
-21 °C	~ -30 °C	C5
-31 °C	~ -40 °C	C6
-41 °C	~ -50 °C	C7

Fluid Temperature		Personnel Protection Insulation Symbol
Below	100 °C	P
101 °C	~ 150 °C	P2
151 °C	~ 200 °C	P3
201 °C	~ 250 °C	P4
251 °C	~ 300 °C	P5
301 °C	~ 350 °C	P6
351 °C	~ 400 °C	P7
401 °C	~ 450 °C	P8
451 °C	~ 500 °C	P9
501 °C	~ 550 °C	P10